

CHE213: MASS TRANSFER INNOVATION PROJECT

GROUP 16



OUR TEAM

ANSHU 240147

HARSH 240433

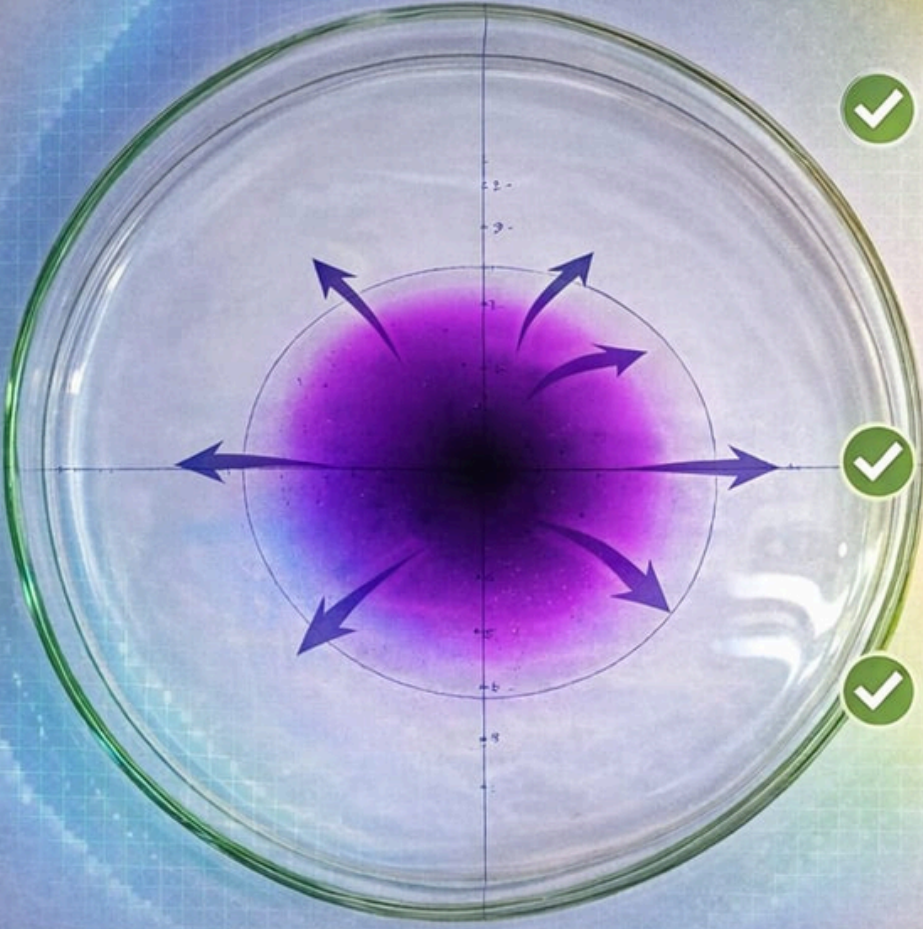
TENZIN 241099

TRIPTI 241103

Objective

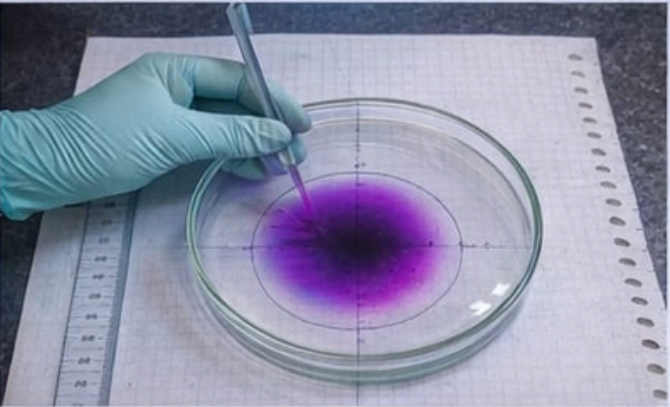
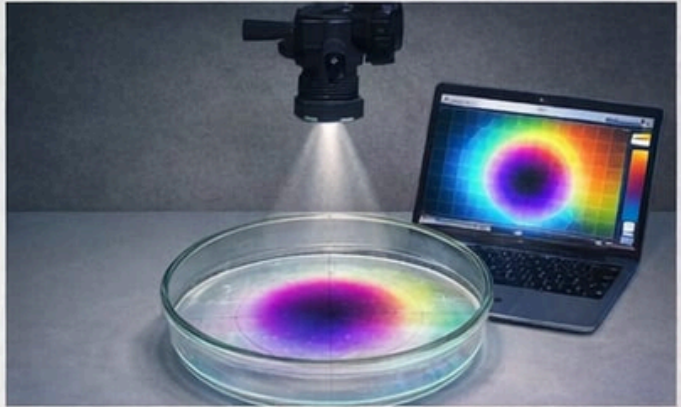
- To perform a comprehensive experimental, analytical, and study of radial diffusion within a stagnant liquid.
- To determine the diffusion coefficient (D) of KMnO_4 in Water and Water-Glycerol mixture using image intensity analysis.
- To analyse how variables such as temperature, liquid viscosity, and initial concentrations affect the rate of diffusion.

Traditional sampling causes convection, disturbing stagnant conditions



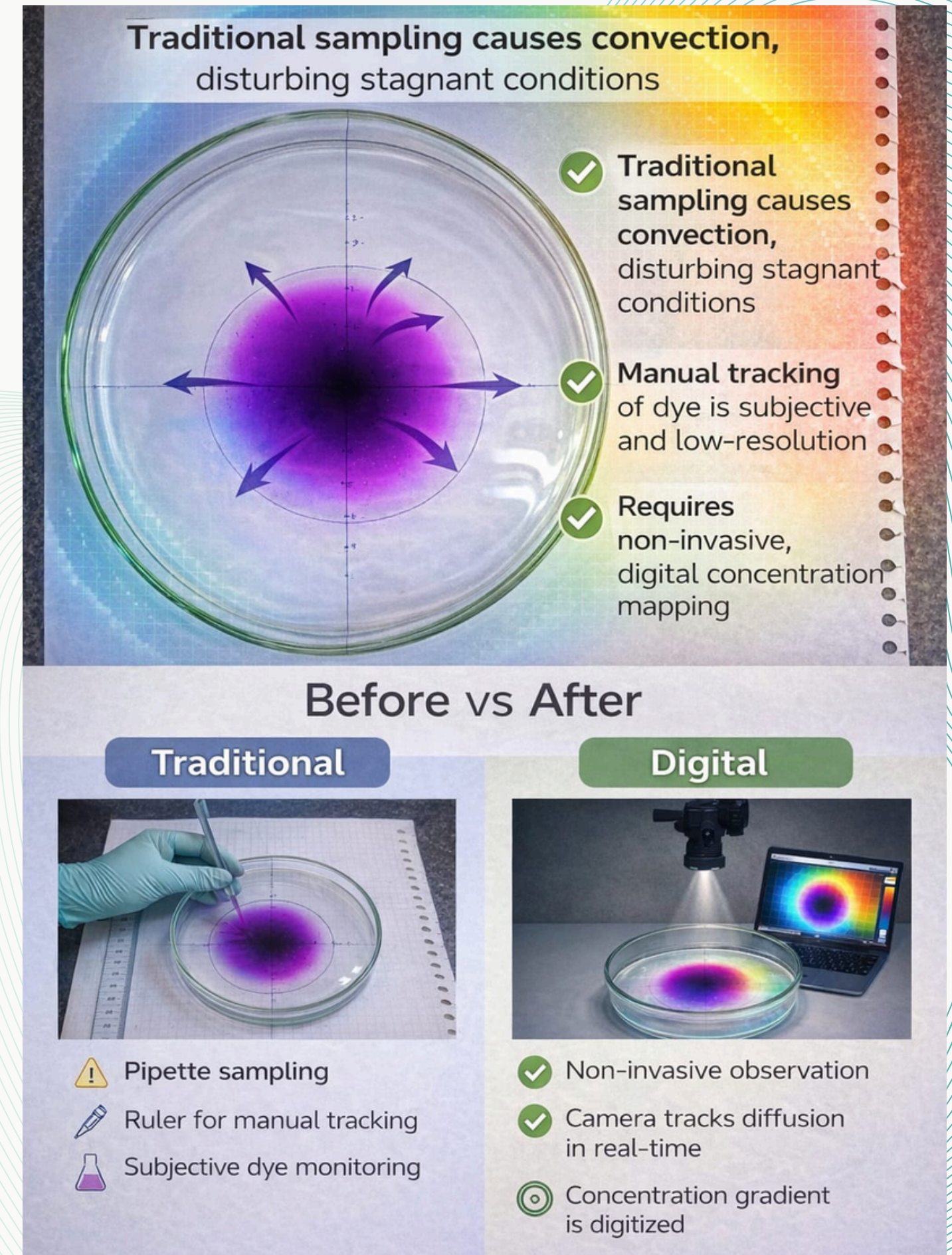
- ✓ Traditional sampling causes convection, disturbing stagnant conditions
- ✓ Manual tracking of dye is subjective and low-resolution
- ✓ Requires non-invasive, digital concentration mapping

Before vs After

Traditional	Digital
	
⚠ Pipette sampling	✓ Non-invasive observation
📏 Ruler for manual tracking	✓ Camera tracks diffusion in real-time
🧪 Subjective dye monitoring	🎯 Concentration gradient is digitized

- **Complex effect of parameters:**

Studying how factors like temperature and viscosity (using water–glycerol mixtures) affect diffusion is difficult with simple methods. A more reliable, non-invasive, and digital approach is needed to measure concentration changes continuously and accurately.



Working principle

- A drop of KMnO_4 in a stagnant liquid spreads radially outward due to the concentration gradient in a petri dish.
- This movement is governed by Fick's Second Law in cylindrical coordinates:

$$\partial C / \partial t = D \left(\partial^2 C / \partial r^2 + (1/r) \partial C / \partial r \right)$$

- Video of KMnO_4 diffusion is processed using MATLAB/Python to convert pixel intensity into concentration profiles, assuming $C = kI$
- Diffusion coefficient is determined using $r^2 \propto Dt$ and validated by comparison with a numerical model
- Experimental results are validated against a numerical finite difference model to predict concentration profiles C_i^{n+1}

Calculations and Derivation

The governing Partial Differential Equation (PDE) is:

$$\frac{\partial C}{\partial t} = D \left(\frac{\partial^2 C}{\partial r^2} + \frac{1}{r} \frac{\partial C}{\partial r} \right)$$

Boundary and Initial Conditions

To solve the PDE, we define the following constraints:

- **Initial Condition** ($t = 0$): All mass M is concentrated at $r = 0$, $C = C_0$.
- **Boundary Condition 1** ($r \rightarrow \infty$): $C(\infty, t) = 0$ (Concentration vanishes at infinite distance).
- **Boundary Condition 2** ($r = 0$): $\frac{\partial C}{\partial r} = 0$ (Symmetry condition at the center).

Calculations and Derivation

Analytical Solution via Similarity Variable

We introduce a similarity variable η , which combines r and t :

$$\eta = \frac{r}{\sqrt{4Dt}}$$

By transforming the PDE into an Ordinary Differential Equation (ODE) and integrating over the total mass M , we obtain the Gaussian concentration profile:

$$C(r, t) = \frac{M}{4\pi Dt} \exp\left(-\frac{r^2}{4Dt}\right)$$

Physical Assumptions

1. **Radial Symmetry:** The concentration C varies only with the radius r and time t .
2. **Constant Diffusivity:** The diffusion coefficient D is independent of concentration.
3. **Infinite Medium:** The dimensions of the petri dish are large enough that the dye front does not reach the edges during the observation period.
4. **Conservation of Mass:** A finite mass M of dye is injected as a point source at $t = 0$.

Calculations

As we are using image processing for our concentration here is the relation between intensity and concentration

$$I \propto C$$
$$C = kI$$

Where:

- (I) = image pixel intensity
- (C) = concentration

$$\frac{C}{C_0} = \frac{I}{I_{max}}$$

Steps:

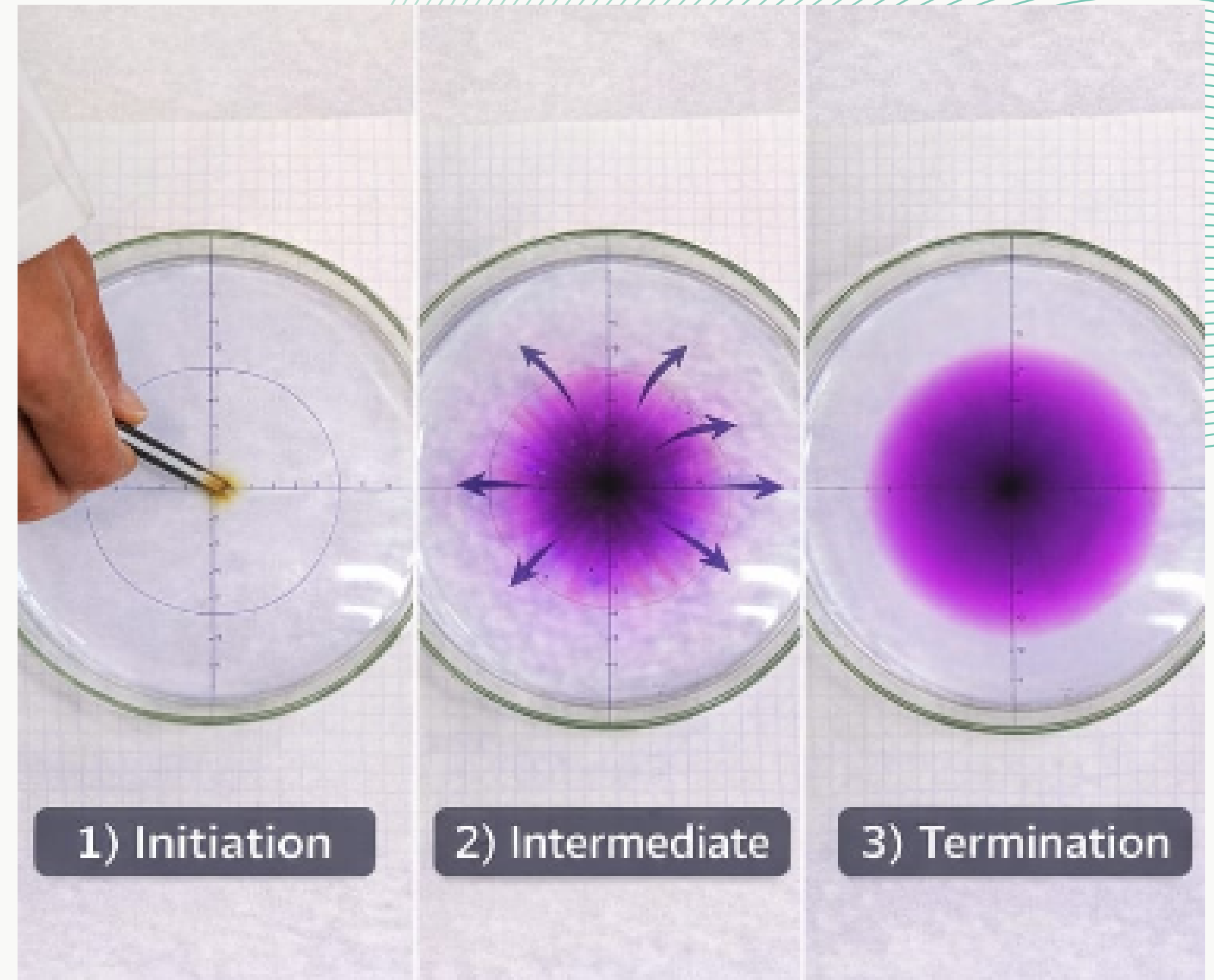
1. **Convert image to Greyscale.**
2. **Extract intensity along a line from the center**
3. **Normalize intensity**
4. **Obtain Concentration vs radius data.**

EXPERIMENTAL SETUP

1. **Apparatus: 20cm diameter petri dish**
2. **KMnO₄**
3. **For stagnant liquid we need distilled water and glycerol**
4. **Syringe, spatula and pair of tweezers**
5. **Leveler, beakers**
6. **Measuring cylinder**
7. **Tripod with camera**

Experimental Methodology

- The petri dish was placed over a reference scale for measurement.
- Liquids used were distilled water and a glycerol–water mixture.
- Experiments were conducted at three temperatures: cold (10–15°C), room temperature, and hot (50–60°C).
- A small drop of KMnO_4 was introduced at the centre of the liquid.
- The diffusion process was recorded using a high-resolution camera.
- Images were captured at regular time intervals.
- The radius of the diffusion front was measured from the recorded images.
- The experiment was repeated for all liquid and temperature combinations.
- The collected data were used for further analysis of diffusion behaviour.



Code Implementation

1. Global Intensity Calibration

```
% Reference intensity from the first drop to anchor C0
globalDarkest = min(firstFrame(:));
globalLightest = max(firstFrame(:));

% Global Normalization formula
C_exp = (globalLightest - I) ./ (globalLightest - globalDarkest);
```

Explains how we converted pixel brightness into physical concentration.

2. Solving Fick's Law

```
% Theoretical:  $C = C_0 * \exp(-r^2 / 4Dt)$ 
% Linearized:  $\ln(C) = (-1/4Dt) * r^2$ 
lnC = log(C_exp);
p = polyfit(rSq(valid), lnC(valid), 1);
slope = p(1);
D = abs(-1 / (4 * slope * t_target));
```

Shows we turned exponential curve into simple linear regression to calculate F

Code Implementation

3. Gradient focused Data selection

```
% Ignoring the saturated 'floor' and the noisy 'edge'  
valid = (C_exp > 0.15 & C_exp < 0.9);
```

4. Dynamic Origin Tracking

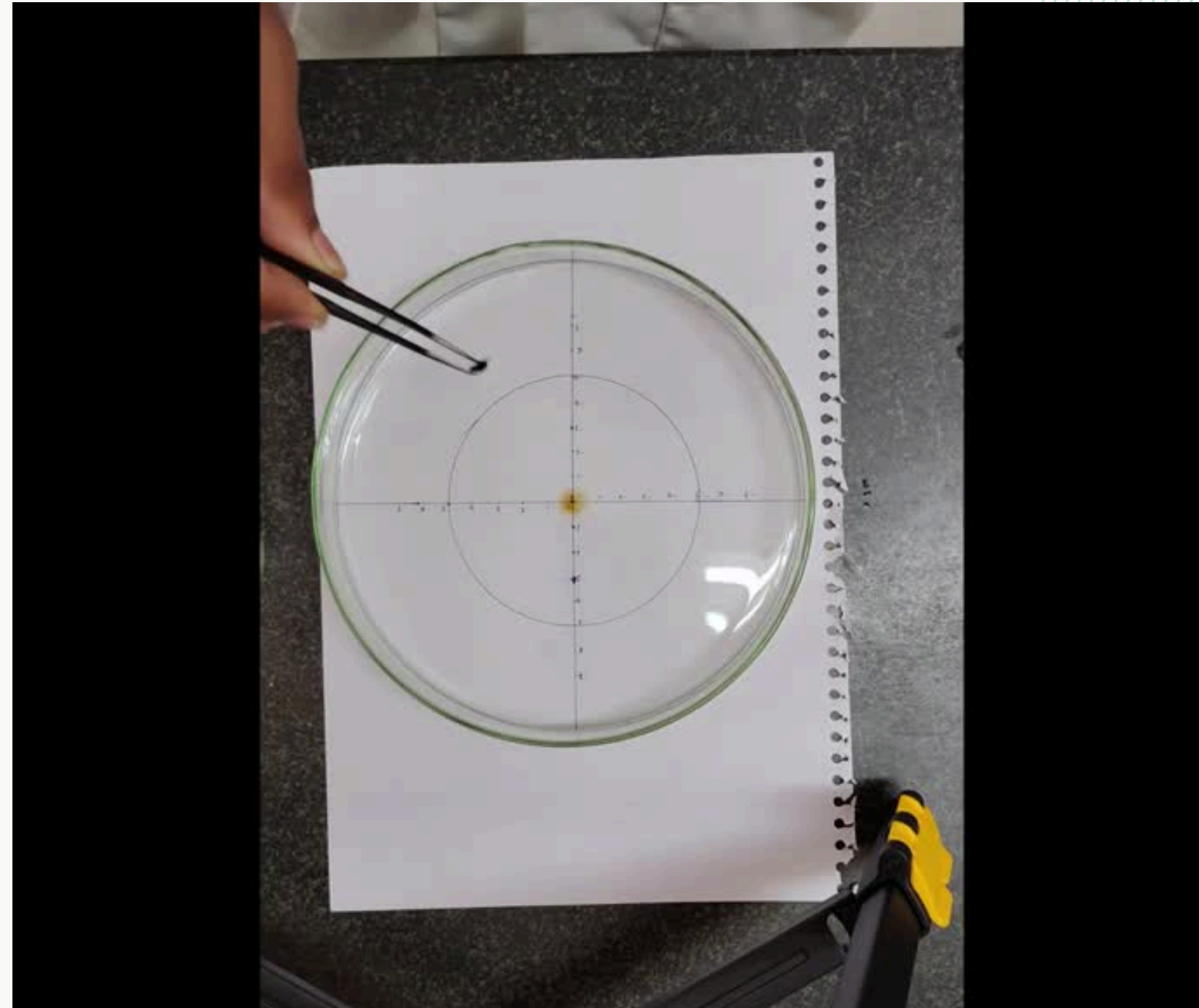
```
% Manual origin recalibration per frame  
title('1. Click CURRENT Center | 2. Click CLEAR Edge');  
[cols, rows] = ginput(2);
```

Shows we accounted for the fact dish or camera might have moved slightly

Experimental Data (Hot Water)

Radius (cm)	Time (min)	r2 (cm2)	Cexp	ln(C)	
	0	0	0	1	0
	0.25	0.06	0.06	0.9961	-0.0039
	0.5	0.25	0.25	0.9804	-0.0198
	0.75	0.57	0.57	0.9674	-0.0331
	1.01	1.01	1.01	0.9569	-0.0441
	1.26	1.58	1.58	0.9451	-0.0565
	1.51	2.27	2.27	0.9333	-0.0691
	1.76	3.09	3.09	0.9216	-0.0817
	2.01	4.04	4.04	0.9059	-0.0988
	2.26	5.11	5.11	0.8824	-0.1252
	2.51	6.31	6.31	0.8667	-0.1431
	2.76	7.64	7.64	0.851	-0.1614
	3.02	9.09	9.09	0.8289	-0.1876
	3.27	10.67	10.67	0.8078	-0.2134
	3.52	12.37	12.37	0.7843	-0.2429
	3.77	14.2	14.2	0.7647	-0.2683
	4.02	16.16	16.16	0.7412	-0.2995
	4.27	18.24	18.24	0.724	-0.323
	4.52	20.45	20.45	0.7102	-0.3422
	4.77	22.79	22.79	0.6892	-0.3723
	5	25	25	0.6245	-0.4708

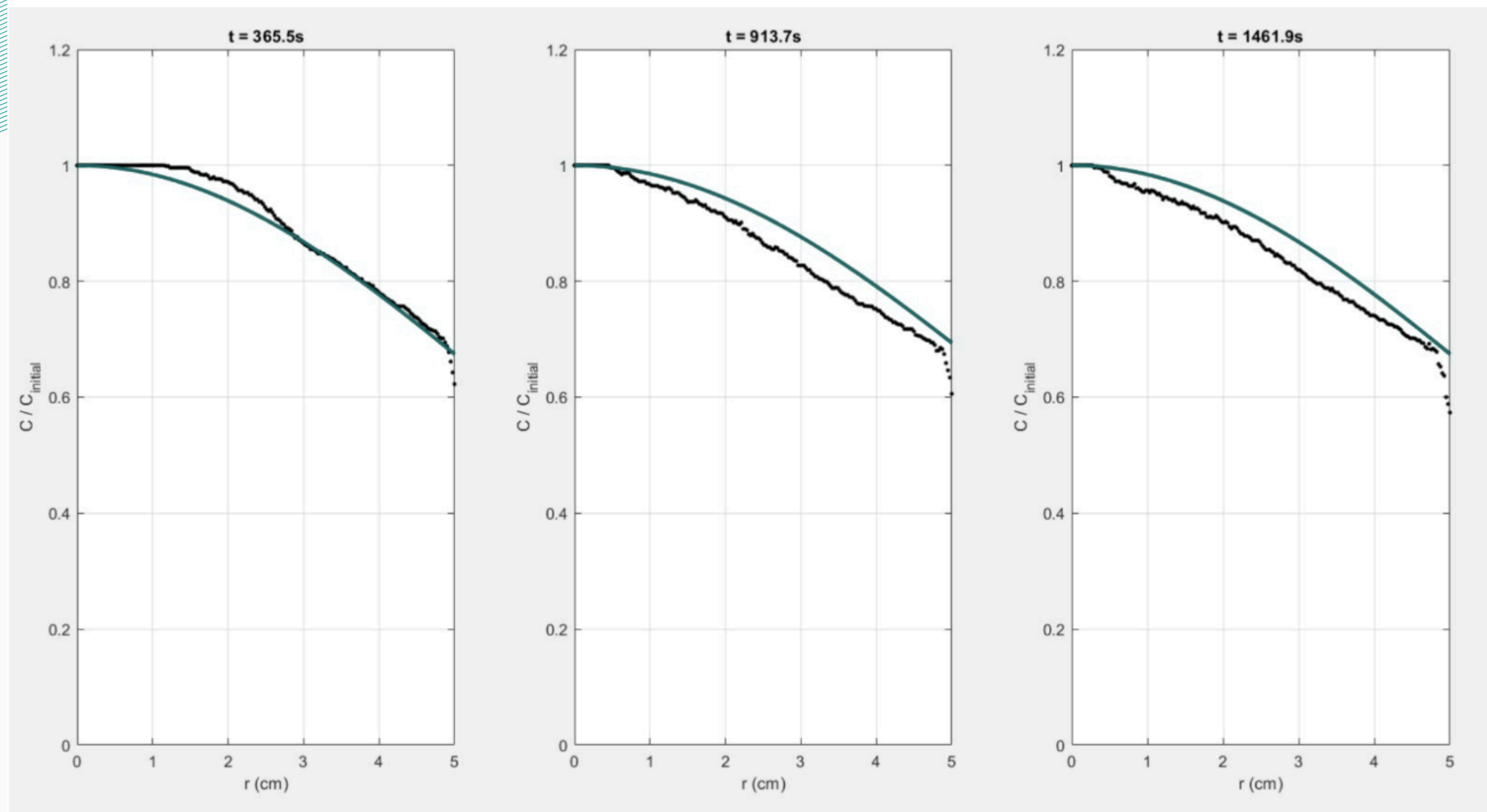
EXPERIMENT VIDEO (HOT WATER)



Time taken approx = 25min

Plots

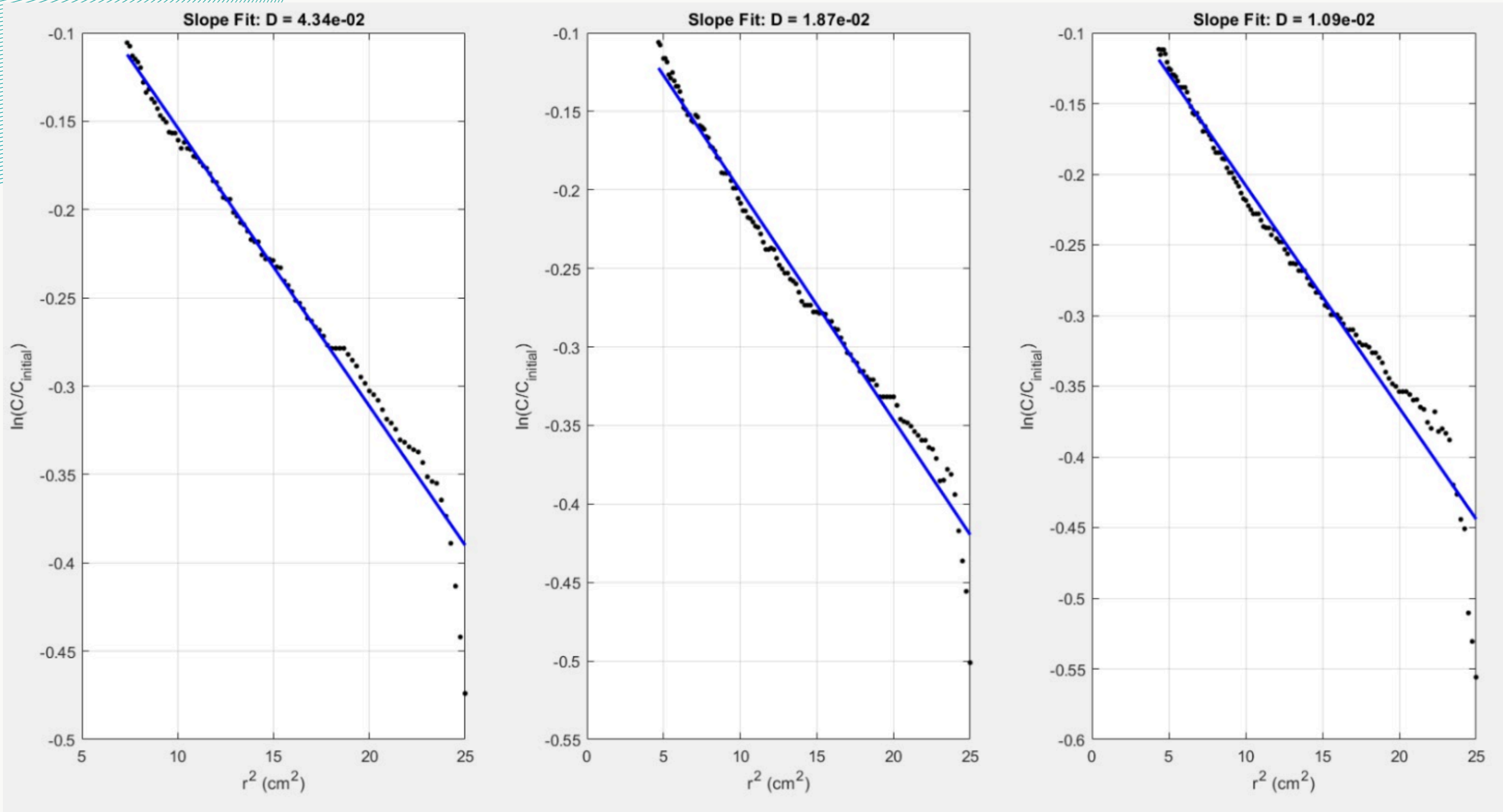
1) Hot Water (50-60°C)



- $D = 2.4335 \times 10^{-2} \text{ cm}^2/\text{s}$

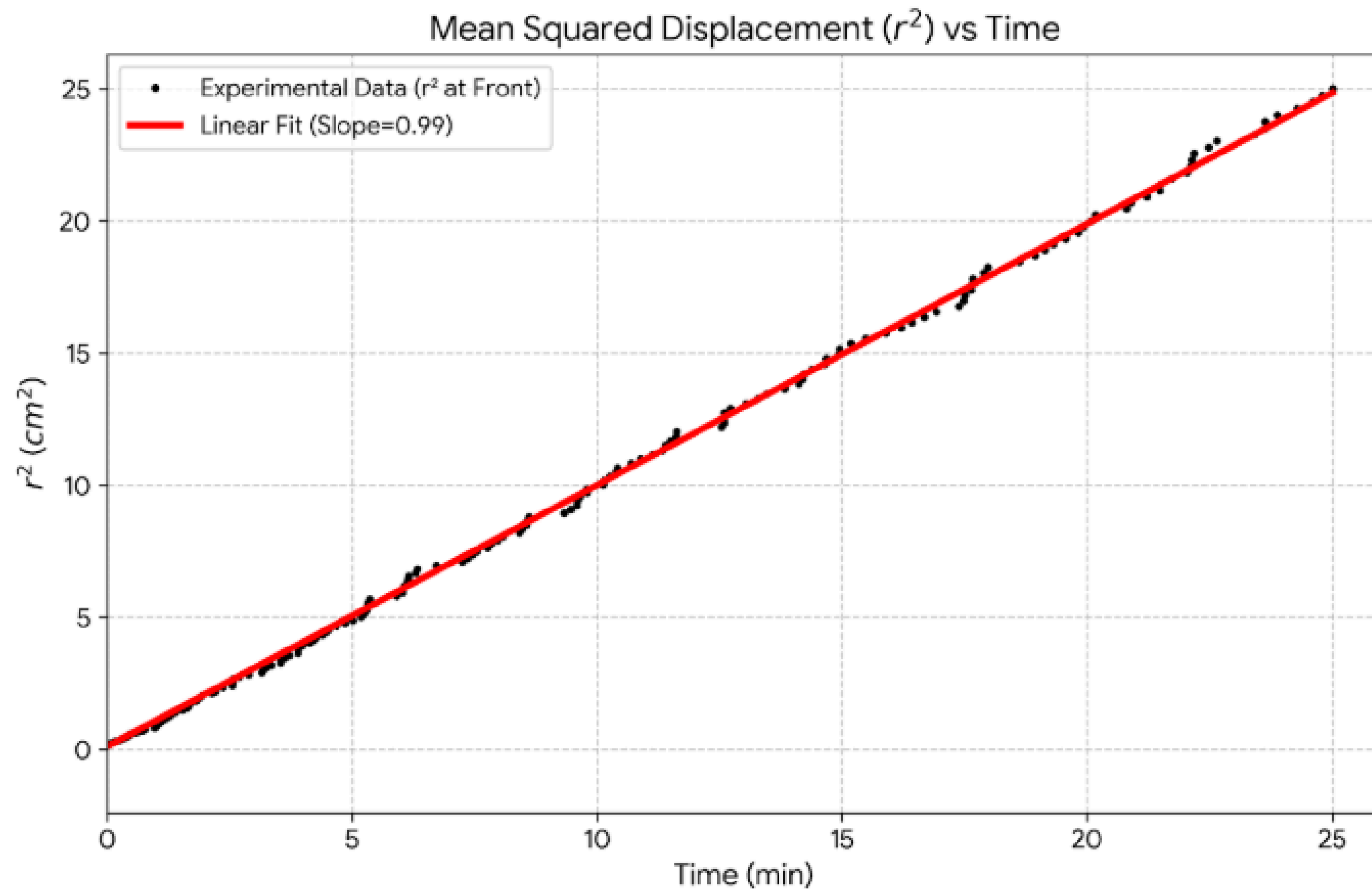
Concentration vs r

1) Hot Water (50-60°C)



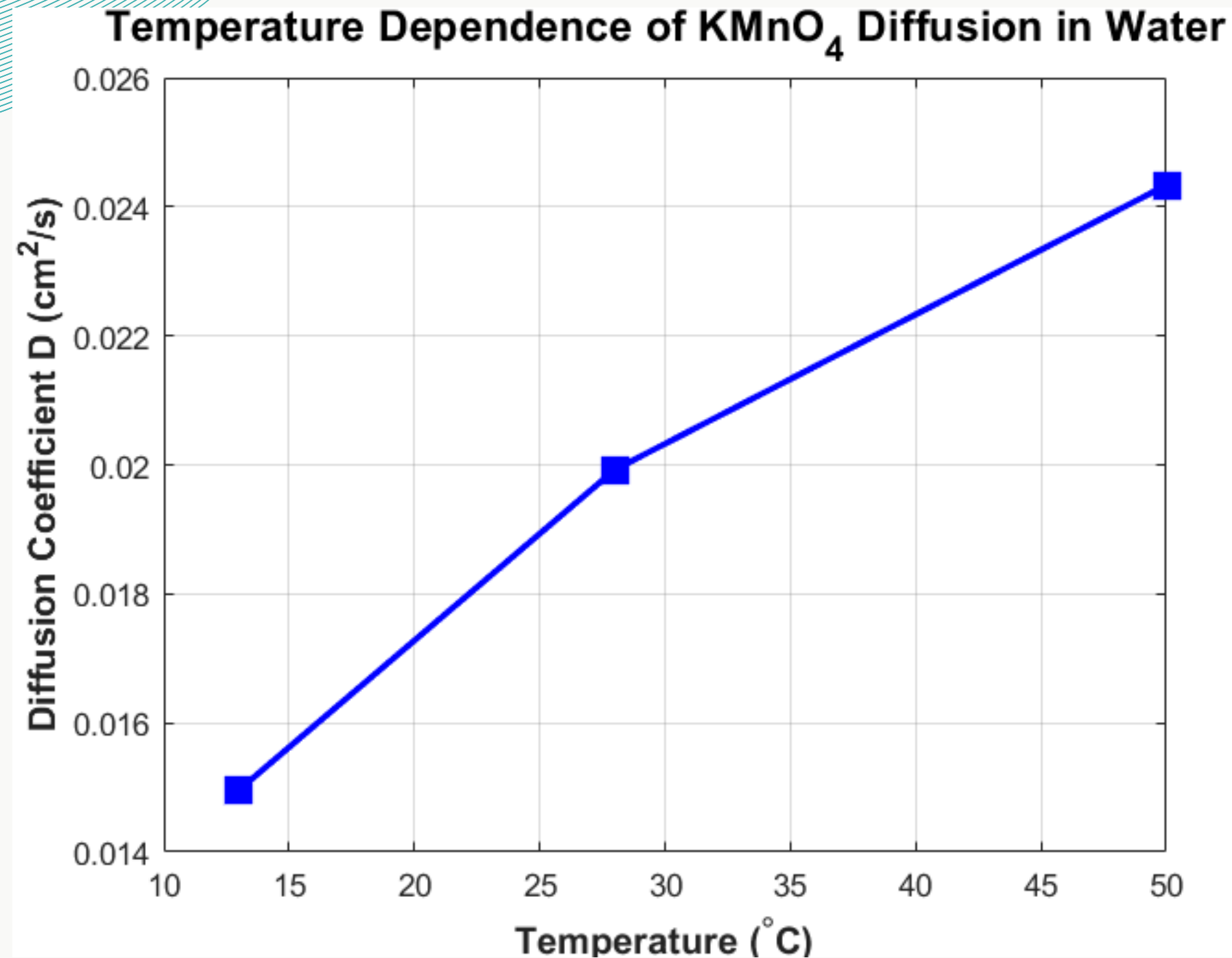
r^2 vs time

1) Hot Water (50-60°C)



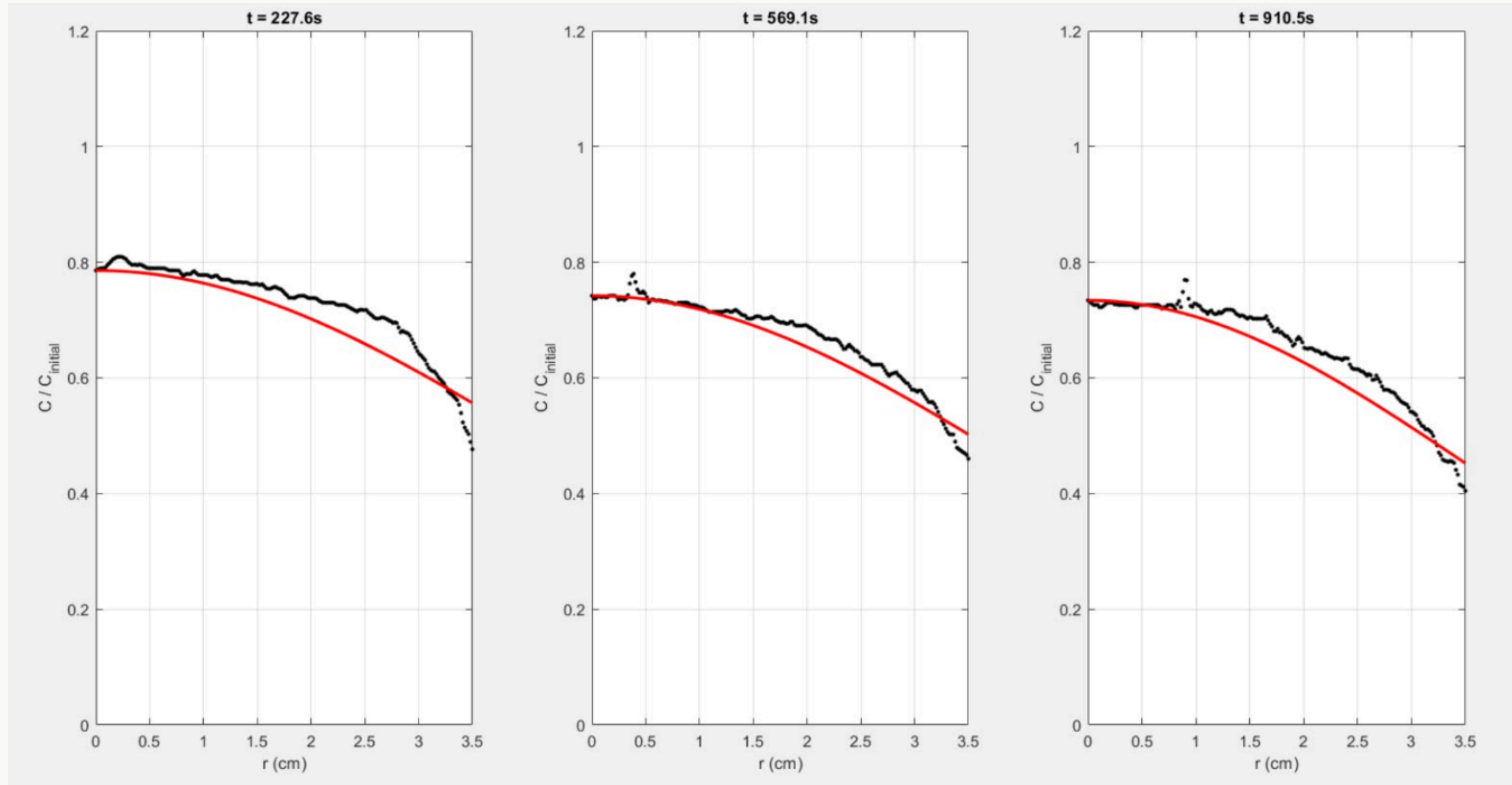
We plotted the r^2 vs time.
We were expected a linear fit of radius square and time.
We can observe that our data points are very near to the linear fit

1) Hot Water (50-60°C)



Here we plotted diffusion coefficient vs Temperature.
we expect diffusion coefficient to increase with temperature and that's what it's showing.

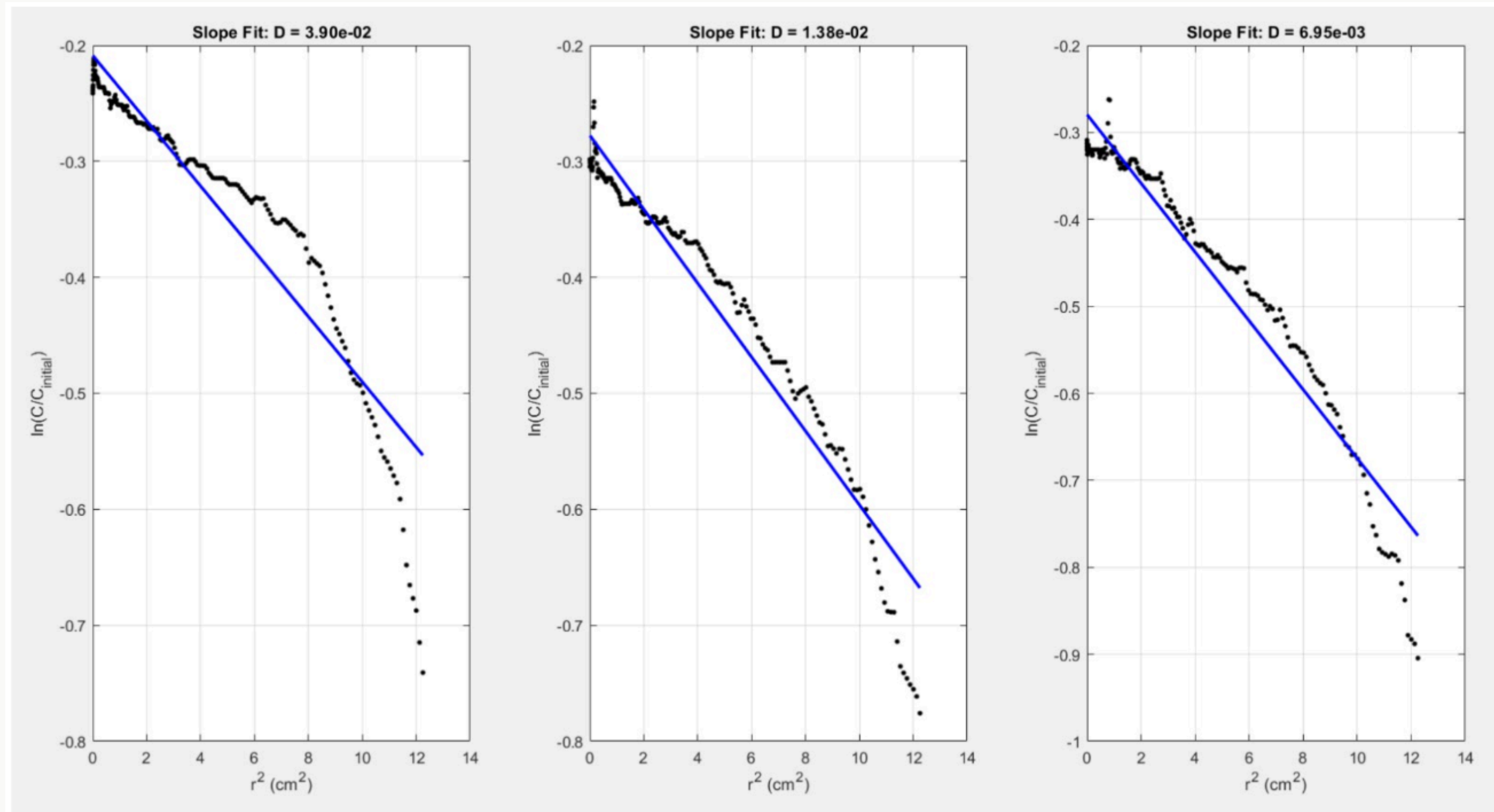
2) Room Temperature Water (28 °C)



Concentration vs. (r)

- $D = 1.9926 \times 10^{-2} \text{ cm}^2/\text{s}$

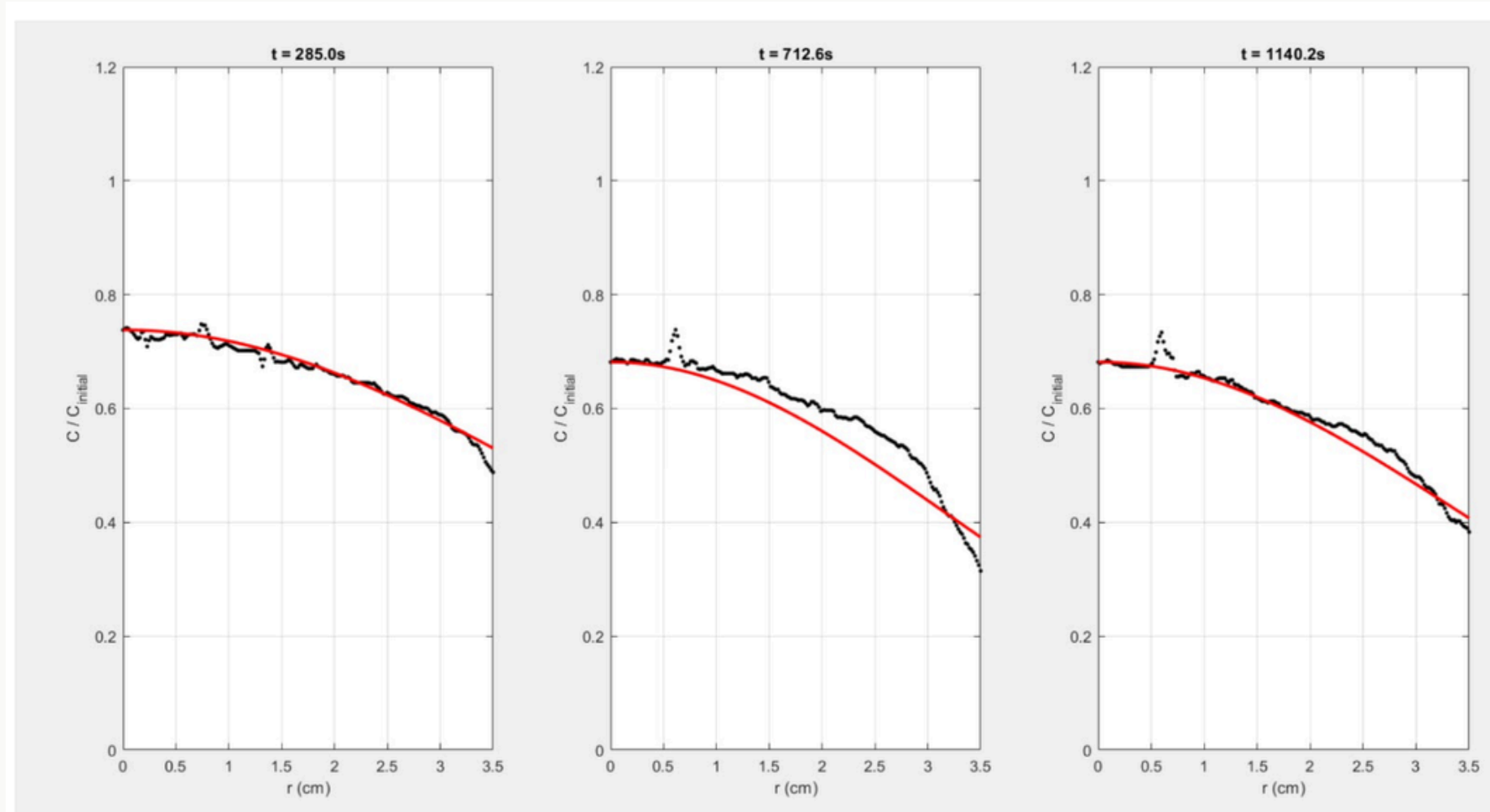
2) Room Temperature Water (28 °C)



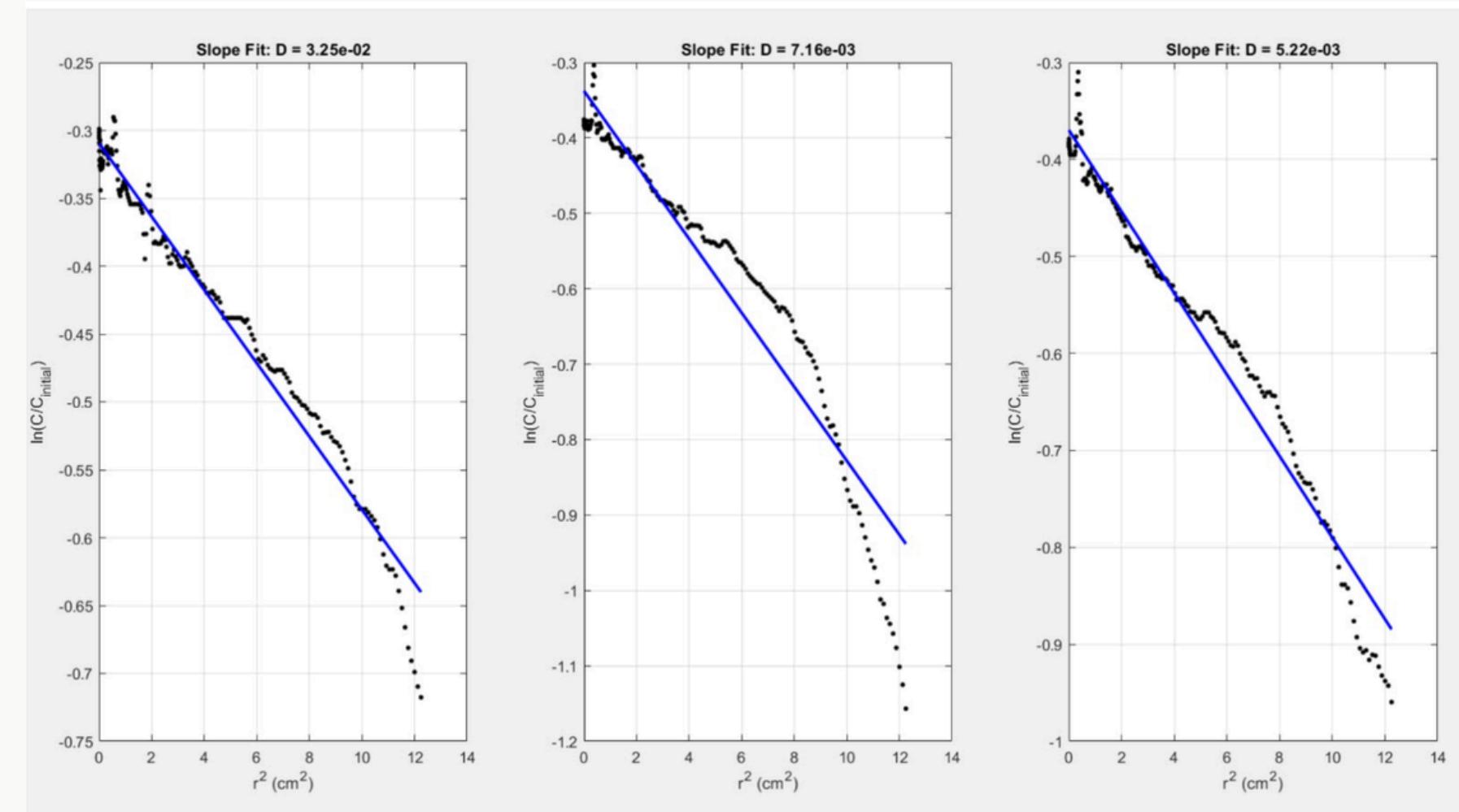
r^2 vs time

- $D = 1.9926\text{e-}02 \text{ cm}^2/\text{s}$

3) Cold Water (10-15°C)



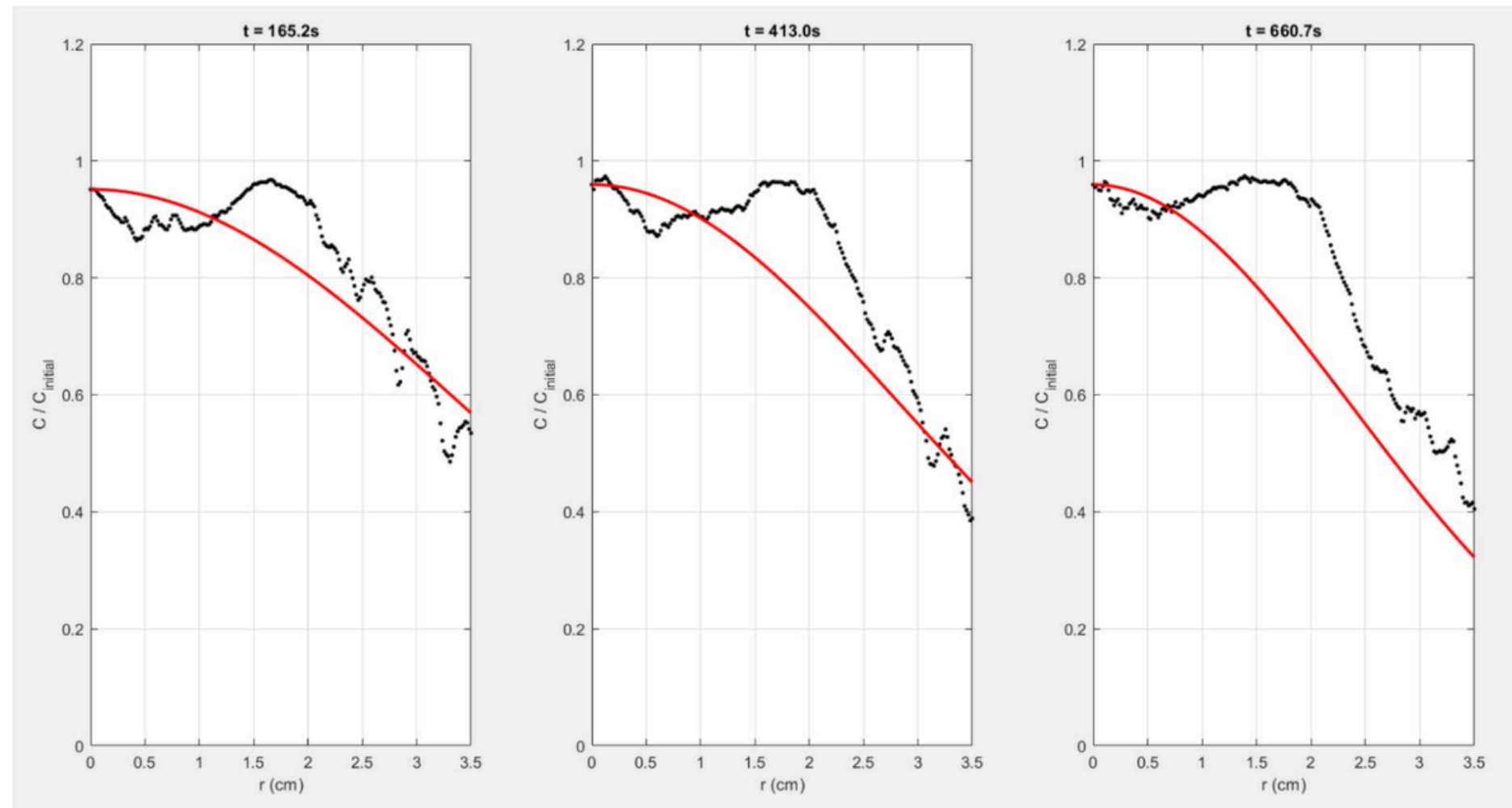
Concentration vs. (r)



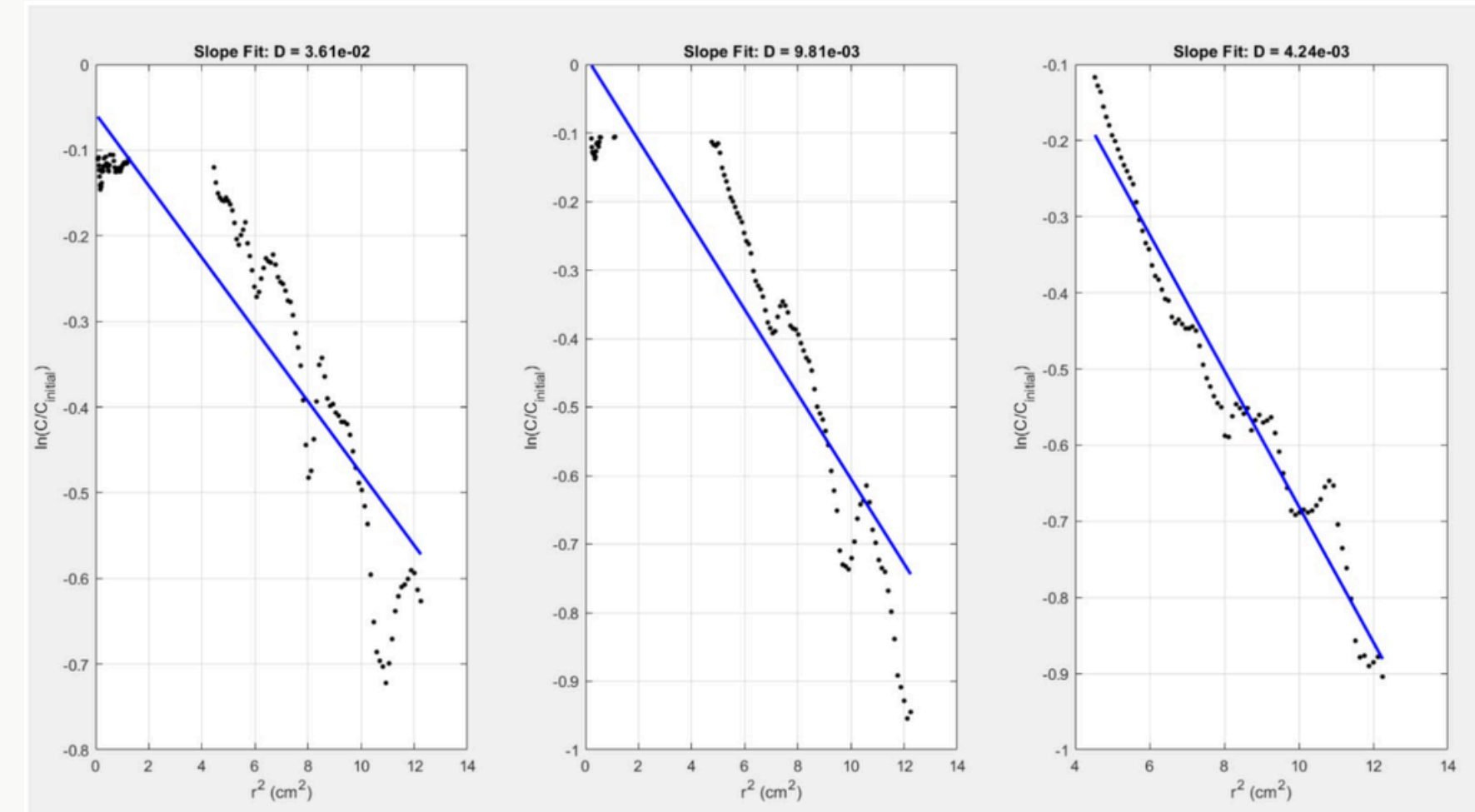
$\ln(C)$ vs r^2

- **$D: 1.4961\text{e-}02 \text{ cm}^2/\text{s}$**

4) Room Temperature Glycerol-Water mixture (28 °C)



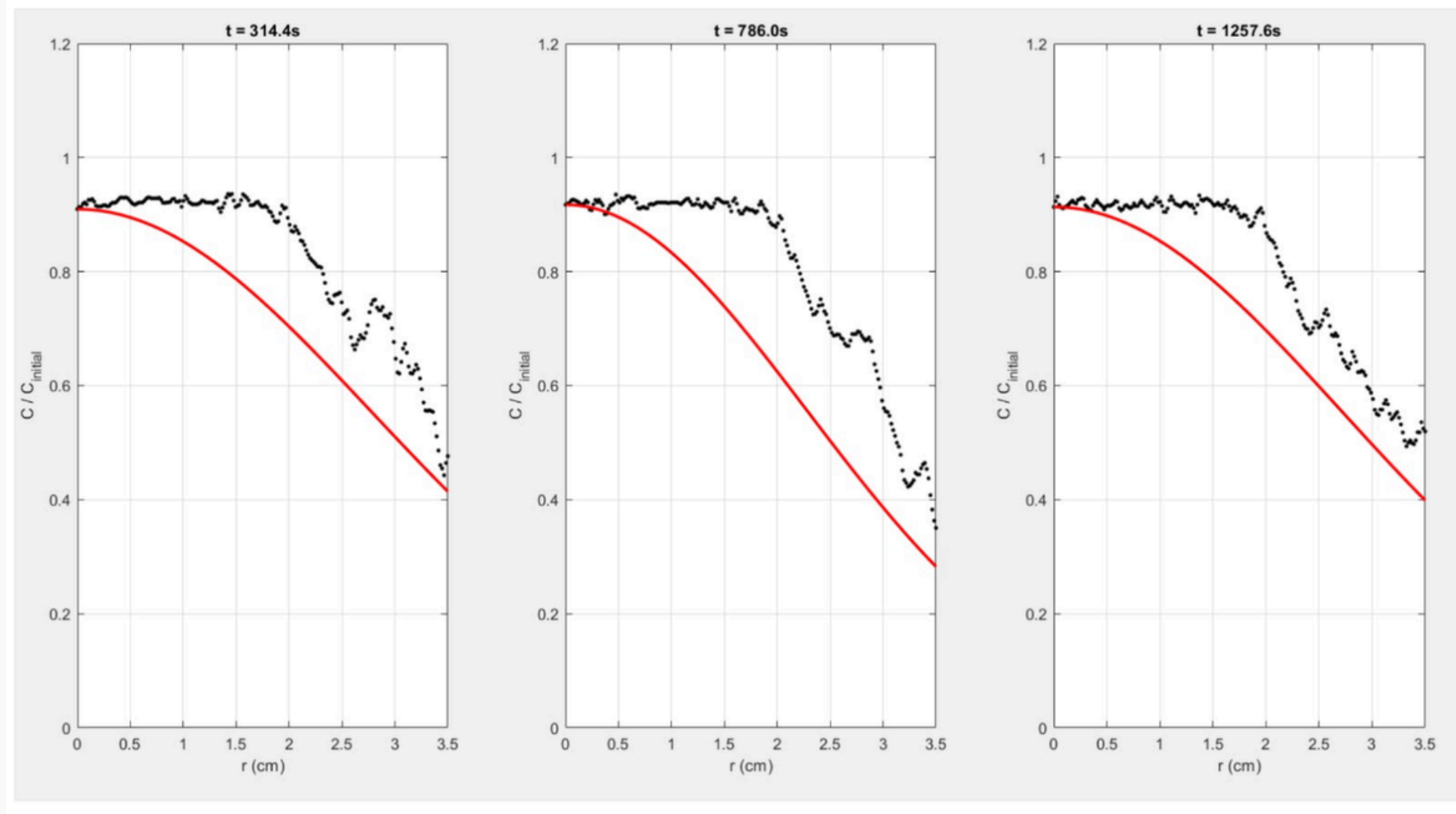
Concentration vs. (r)



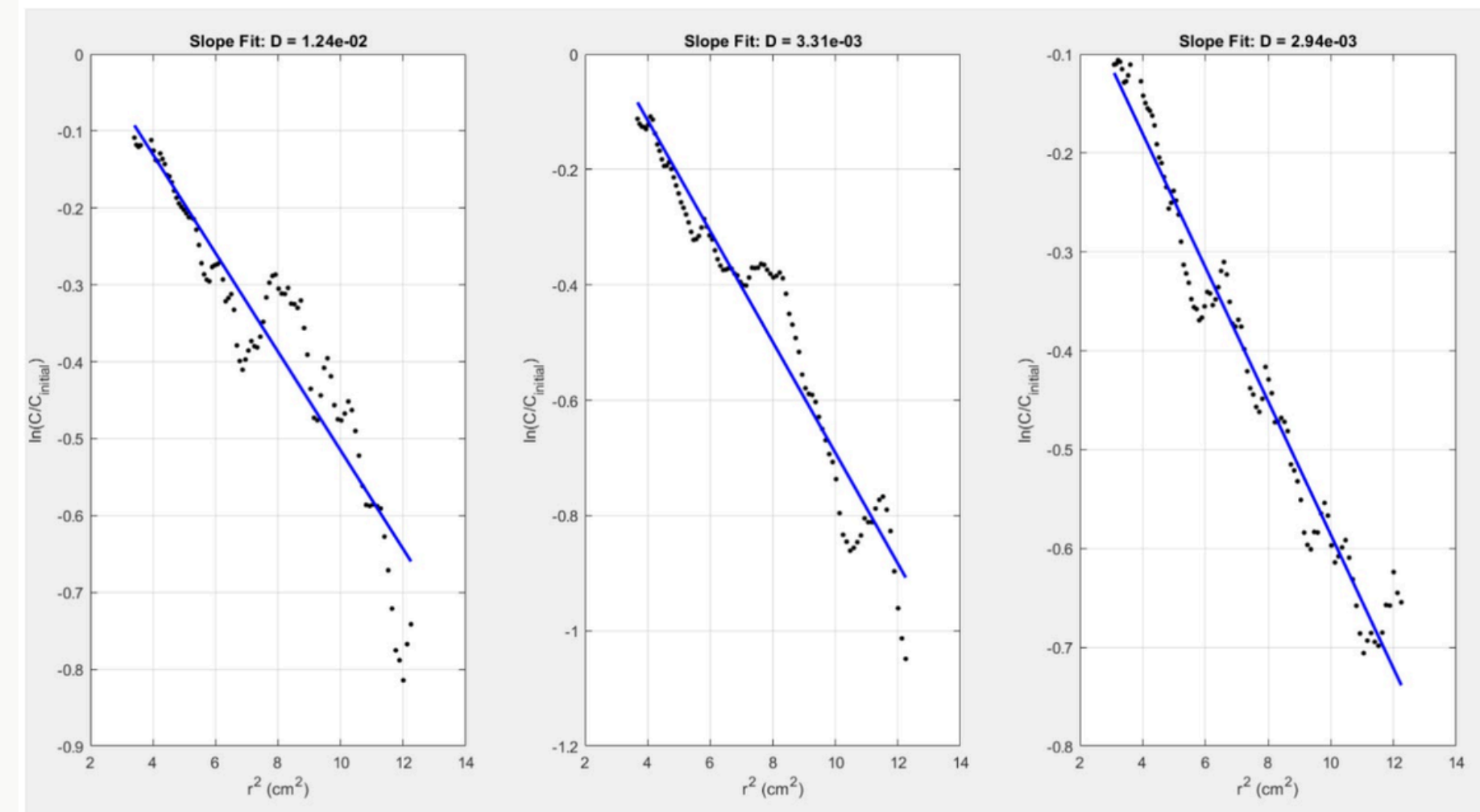
$\ln(C)$ vs r^2

- **$D : 1.6700\text{e-}02 \text{ cm}^2/\text{s}$**

5) Cold Glycerol-Water mixture (10- 15 °C)



Concentration vs. (r)



$\ln(C)$ vs r^2

- **$D: 6.2129\text{e-}03 \text{ cm}^2/\text{s}$**

Observations and Analysis

- Radial Symmetry: All cases showed consistent circular spreading, confirming isotropic diffusion and a level experimental setup.
- Temperature Dependence: D increased significantly from 13°C to 50°C. This proves that higher thermal energy boosts molecular velocity and lowers solvent viscosity, accelerating transport.
- Viscosity Inhibition: The addition of Glycerol drastically slowed expansion. This validates the Stokes-Einstein relation, showing that fluid resistance is a dominant factor in mass transfer.
- Unsteady-State Behavior: Peak concentration at the center decayed over time as the dye redistributed. This "fading" confirms the point-source analytical model.
- Linearity (r^2 vs. time): The linear relationship between squared radius and time across all samples confirms that the transport mechanism was fundamentally Fickian Diffusion

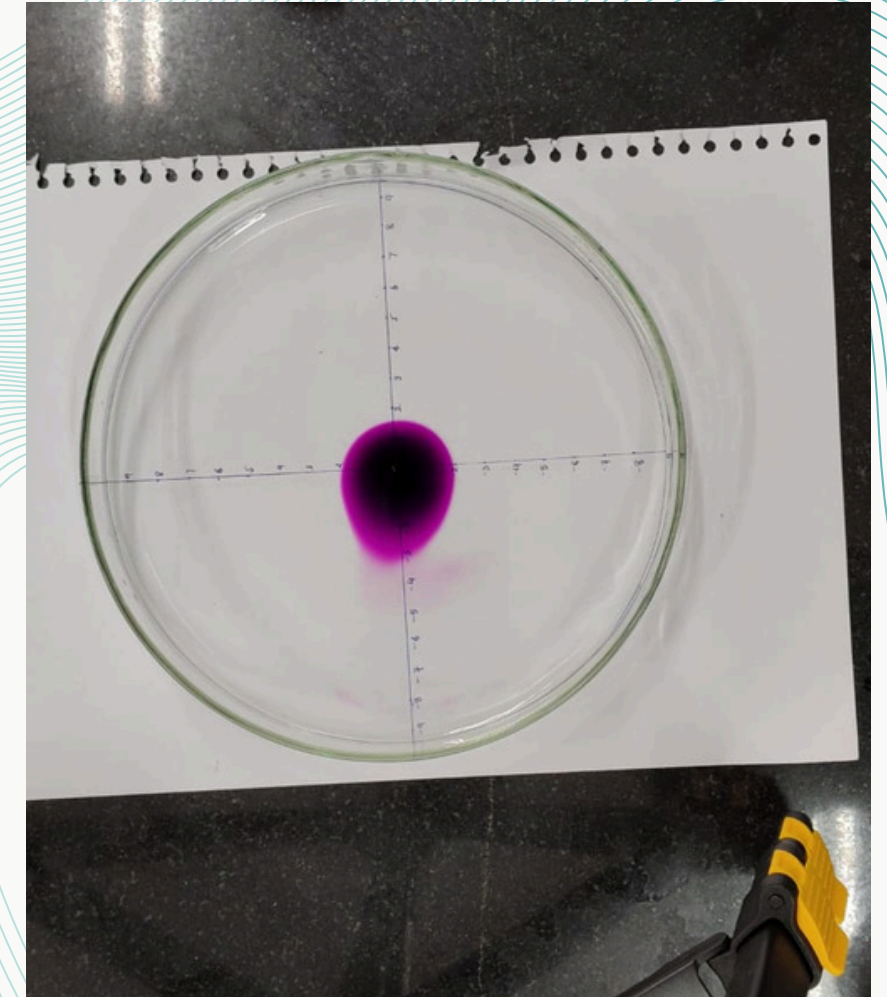
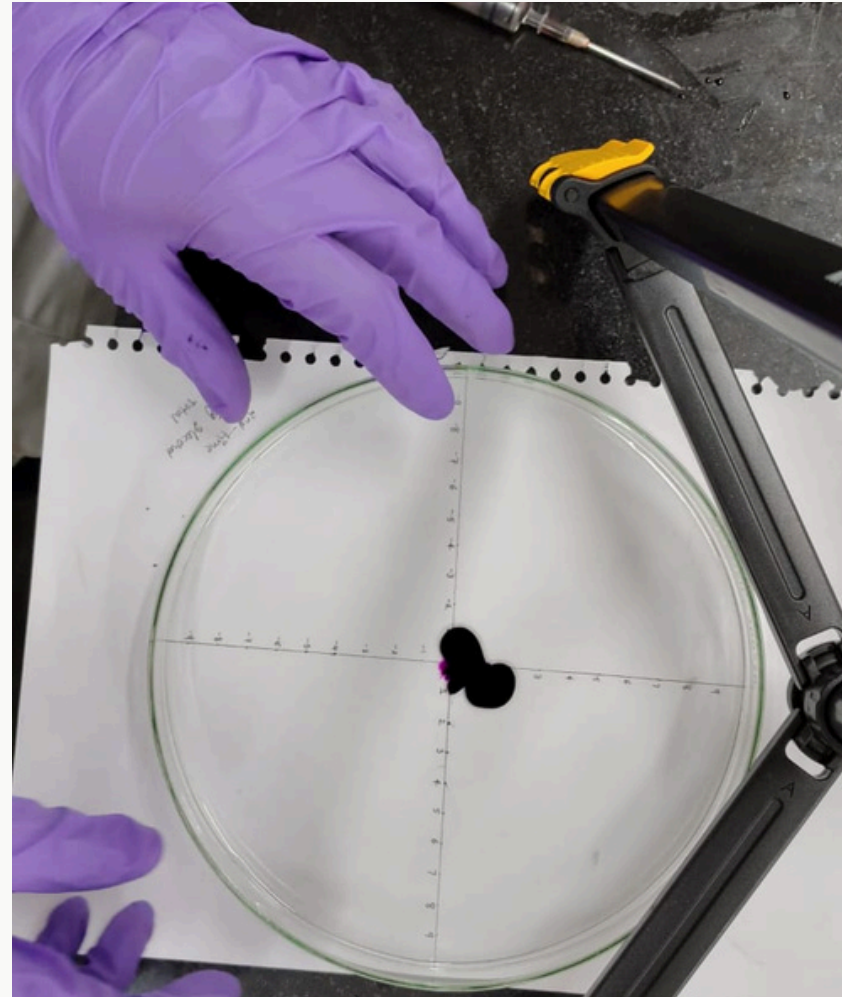
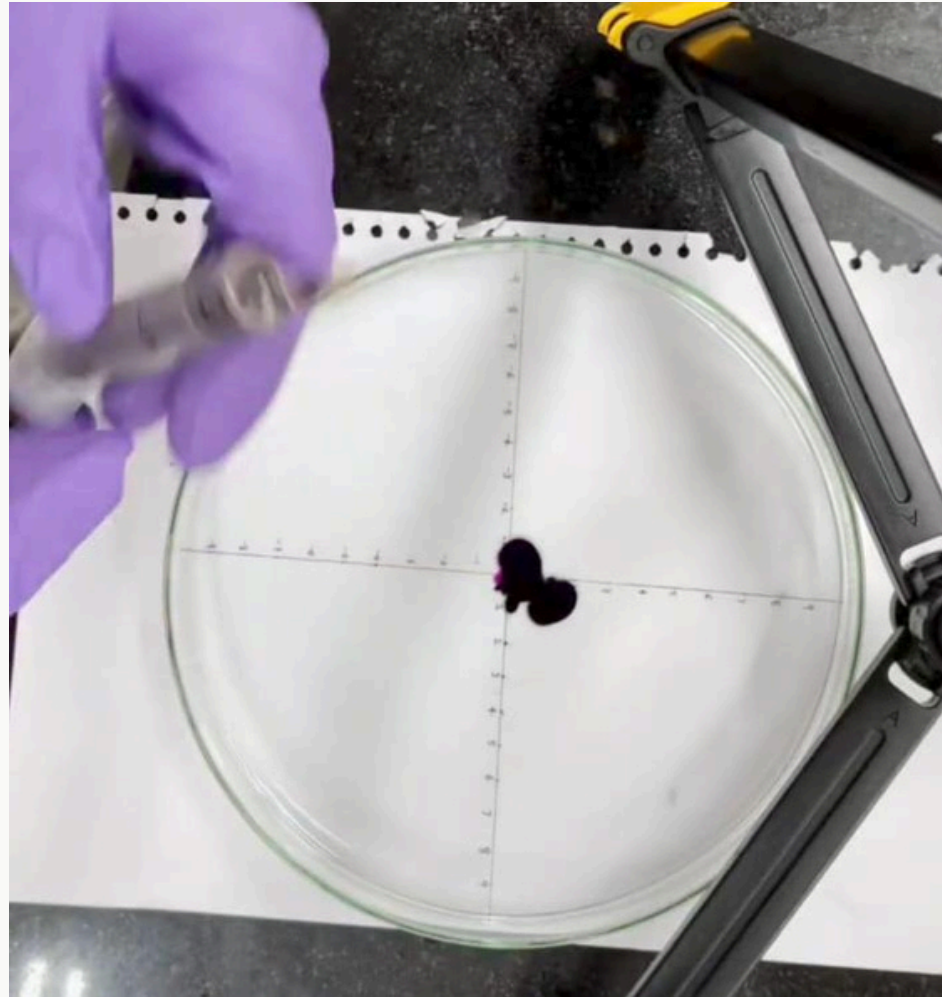
Sources of Error

- Parallax & Camera Alignment: If the camera wasn't perfectly perpendicular to the petri dish, circular dye spreads would appear as ellipses, skewing the radial (r) measurements.
- Initial Momentum: The force of the pipette drop creates a "splash" or mechanical push, causing faster initial spreading that isn't true molecular diffusion.
- Thermal Convection: In the 50°C case, temperature gradients between the hot water and the room-air surface likely created "convection cells," carrying the dye much faster than diffusion alone.
- Density Currents: KMnO_4 is denser than water. The dye "slumps" and spreads across the bottom of the dish due to gravity, leading to higher "apparent" D values.
- Optical Saturation: The camera sensor cannot distinguish between "very dark" and "pitch black." This makes the center of the drop appear wider and flatter (the plateau effect) than it physically is.
- Vibrations: Any movement of the table or floor during the 25-minute runs could induce minor mixing, artificially increasing the spreading rate.
- Glycerol Mixing: If the Glycerol and water weren't 100% homogenized, the dye might have hit "pockets" of different viscosities, causing uneven spreading.

Experimental Precautions

- **Level Surface:** Ensure the petri dish is on a perfectly horizontal surface to prevent gravity-driven flow from biasing the dye spread in one direction.
- **Thermal Equilibrium:** Allow the water/glycerol mix to reach a stable temperature before injecting the dye to avoid unintended internal convection currents.
- **Minimal Drop Height:** Use a micropipette to release the dye as close to the water surface as possible to prevent "splashing" or initial momentum errors.
- **Zero Vibration:** Keep the experimental bench isolated from movements, foot traffic, or loud sounds that could induce mechanical mixing during the long 25-minute runs.
- **Ambient Lighting:** Maintain constant, diffuse lighting to prevent glare on the water surface, which can cause errors in the image processing software.
- **Pure Reagents:** Use distilled water to ensure no impurities or ions interfere with the KMnO_4 diffusion kinetics.
- **Cover the Dish:** Use a transparent lid or cover to prevent evaporation and air currents from disturbing the water surface during long-duration observations.

FAILED ATTEMPTS



Conclusion

The experiment successfully validated the fundamental principles of unsteady-state mass transfer by mapping the radial spreading of KMnO_4 in various media. The results confirm that the diffusion coefficient (D) is a coupled function of thermal activation and medium resistance, consistent with the Stokes-Einstein relation. Higher temperatures significantly accelerated diffusion by reducing solvent viscosity and increasing molecular kinetic energy, while the addition of glycerol acted as a dominant inhibitor. Despite minor experimental noise from convection and density-driven slumping, the high linearity of the r^2 vs. t plots proves that the transport was primarily governed by Fickian kinetics. This study effectively demonstrates the transition from qualitative visual observations to a quantitative analytical model for molecular transport.

THANK

YOU!